DRÆM - ICCV 2021

A Discriminatively Trained Reconstruction Embedding for Surface Anomaly Detection – Vitjan Zavrtanik, Matej Kristan, Danijel Skočaj

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- OOD samples are often rare and diverse
- Collecting data for all possible anomalies is often infeasible
- Data collection gets even harder when pixel-accurate localization is required
- Therefore, methods that only rely on normal data are preferred

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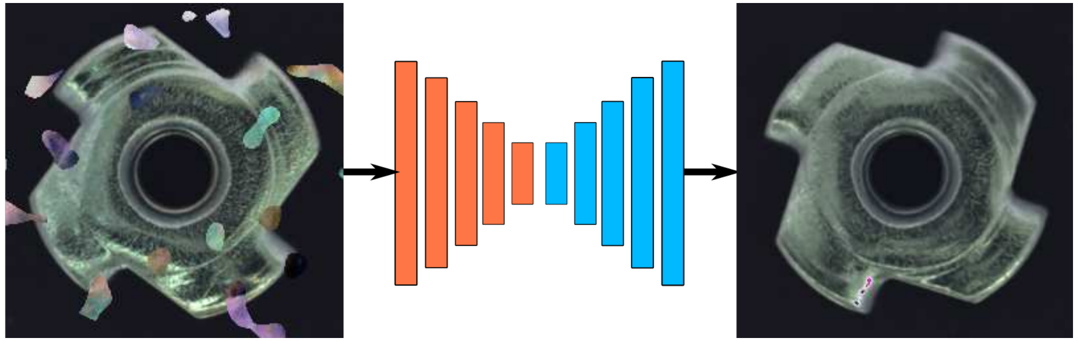

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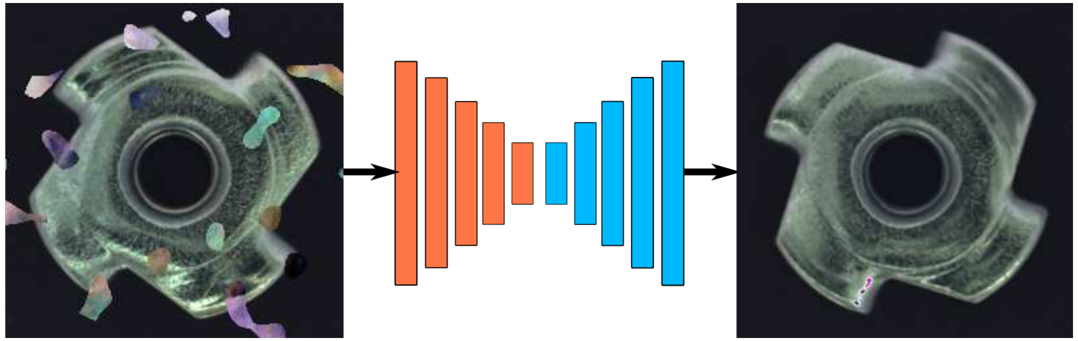

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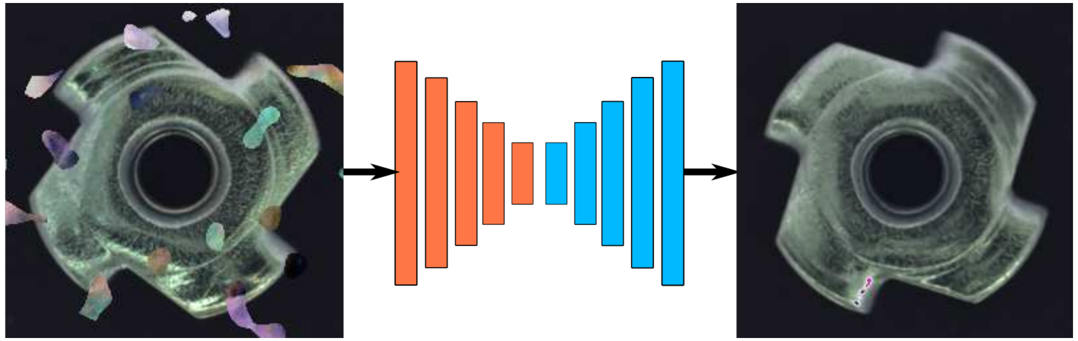

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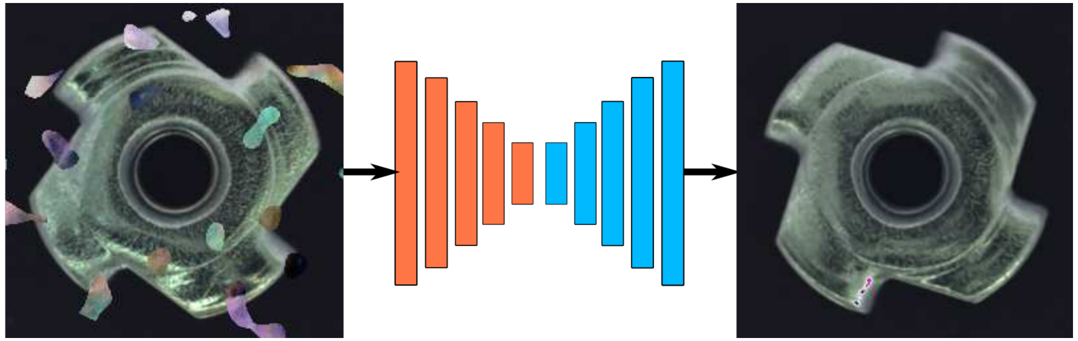
   - Often suffer from blurry or coarse segmentation
   - Overfit to synthetic anomalies



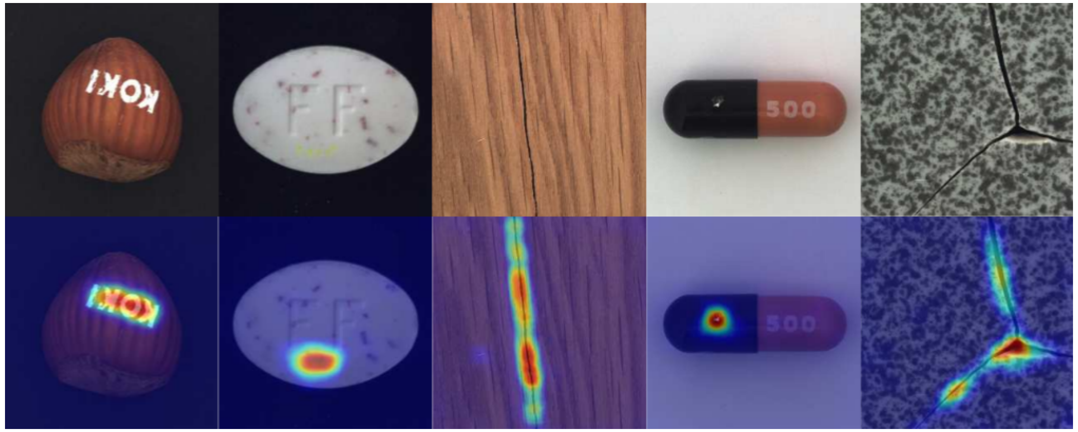
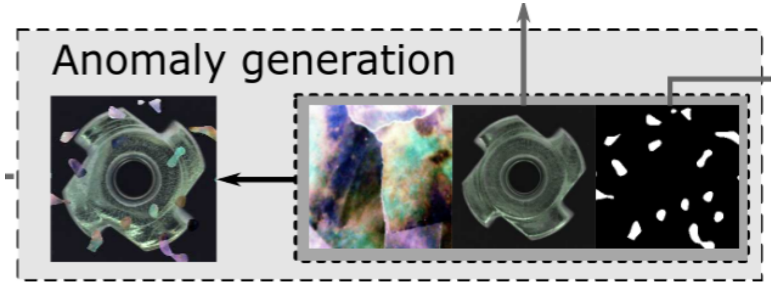


Figure: Blurry segmentation maps from discriminative methods.

Framework Overview
How it Works



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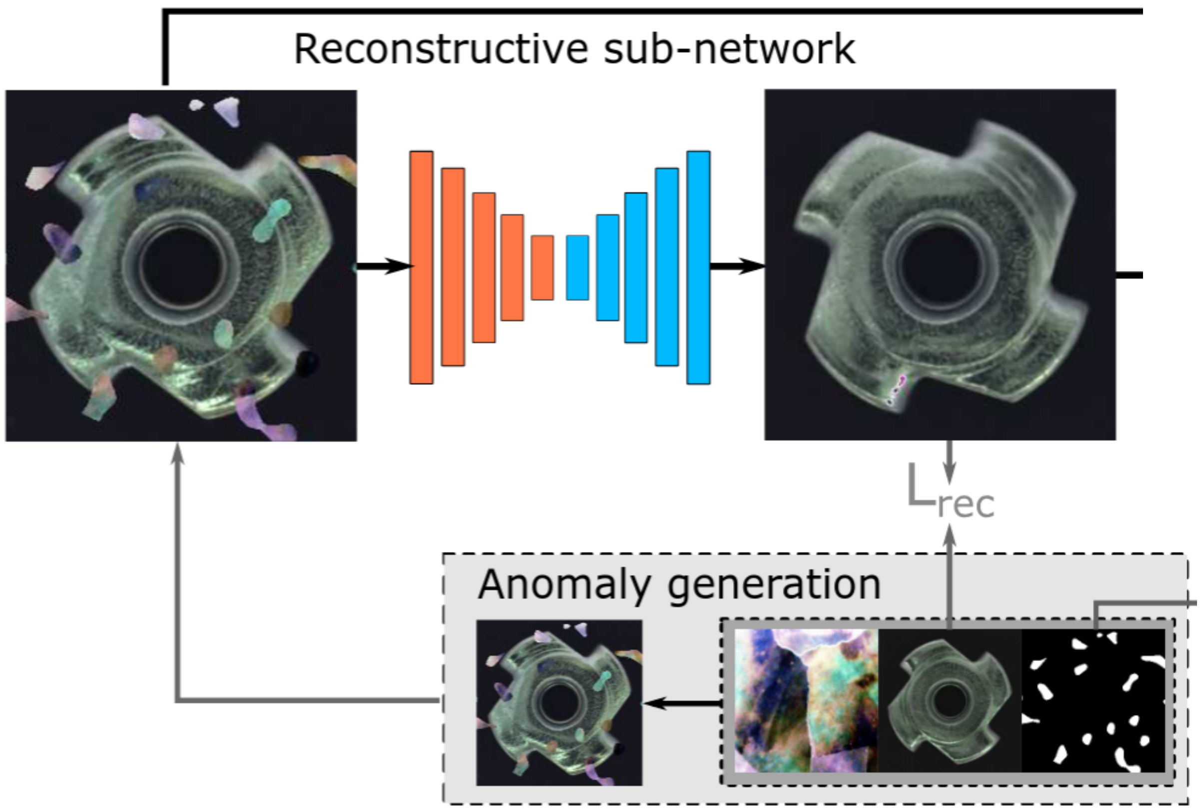

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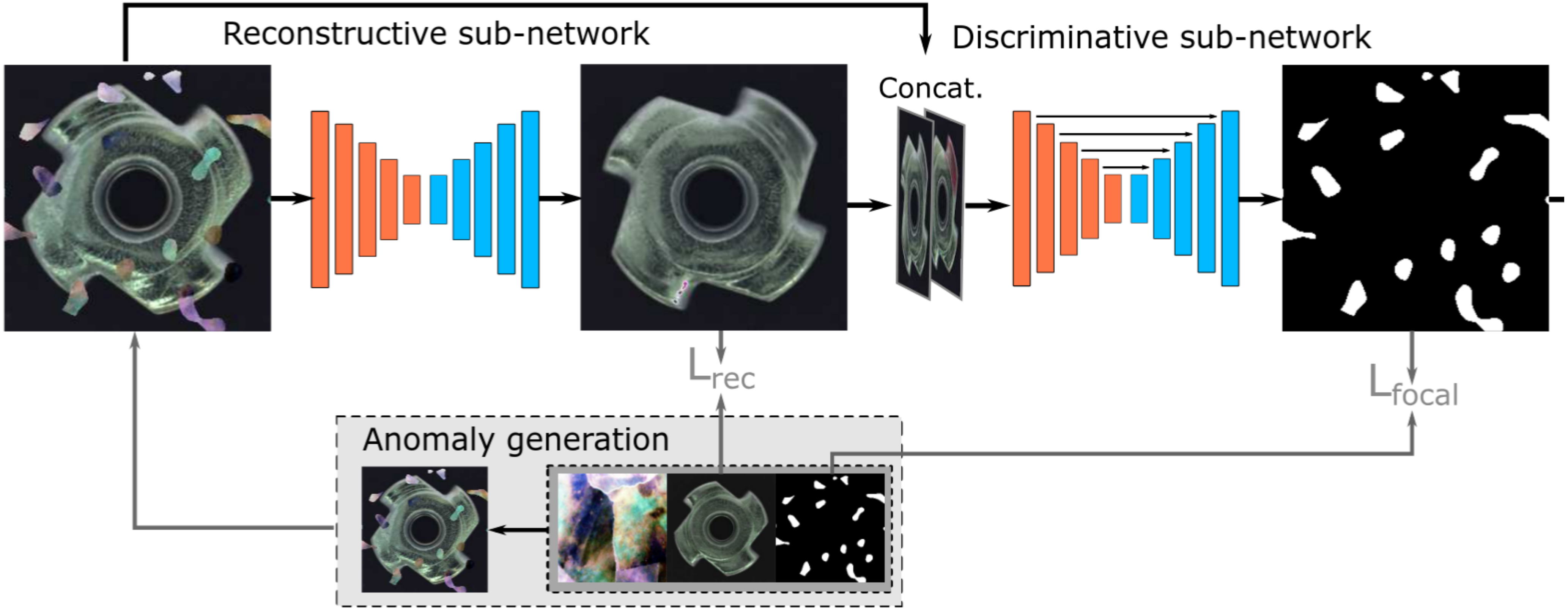

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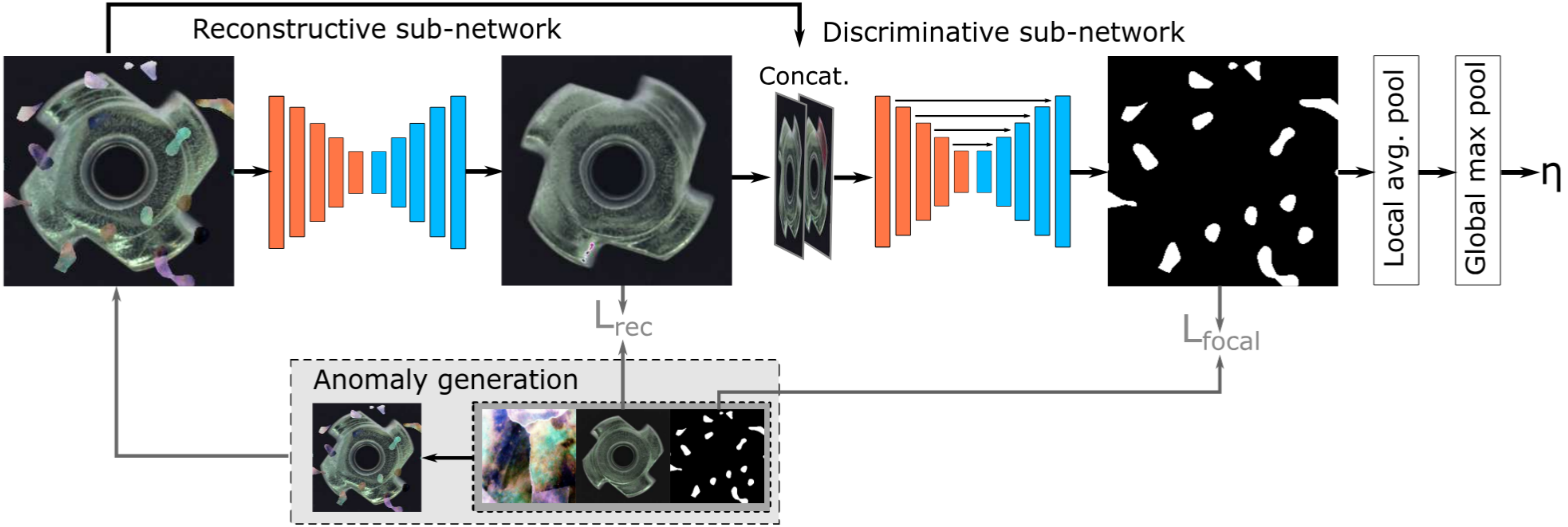

Performance Comparison
DRÆM vs State-of-the-Art

Method	Image AUROC	Pixel AUROC	Pixel AP
US	87.7%	93.9%	45.5%
RIAD	91.7%	94.2%	48.2%
PaDim	<u>95.5%</u>	**97.4%**	<u>55.5%</u>
DRÆM	**98.0%**	<u>97.3%</u>	**68.4%**

Table: Results on the MVTec dataset.

Performance on DAGM Dataset

Unsupervised vs. Supervised Methods

Method	AUROC	TPR	TNR	CA
Unsupervised				
US	72.5	72.6	65.3	66.2
RIAD	78.6	79.2	69.1	70.4
PaDim	95.0	83.3	97.5	95.7
DRÆM	**99.0**	**96.5**	**99.4**	**98.5**

Table: DRÆM on DAGM dataset: outperforming unsupervised methods and on par with fully supervised ones.

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CADN	–	–	–	89.1
Racki et al.	99.6	99.9	99.5	–
Lin et al.	99.0	99.4	99.9	–
Bozic et al.	**100**	**100**	**100**	**100**

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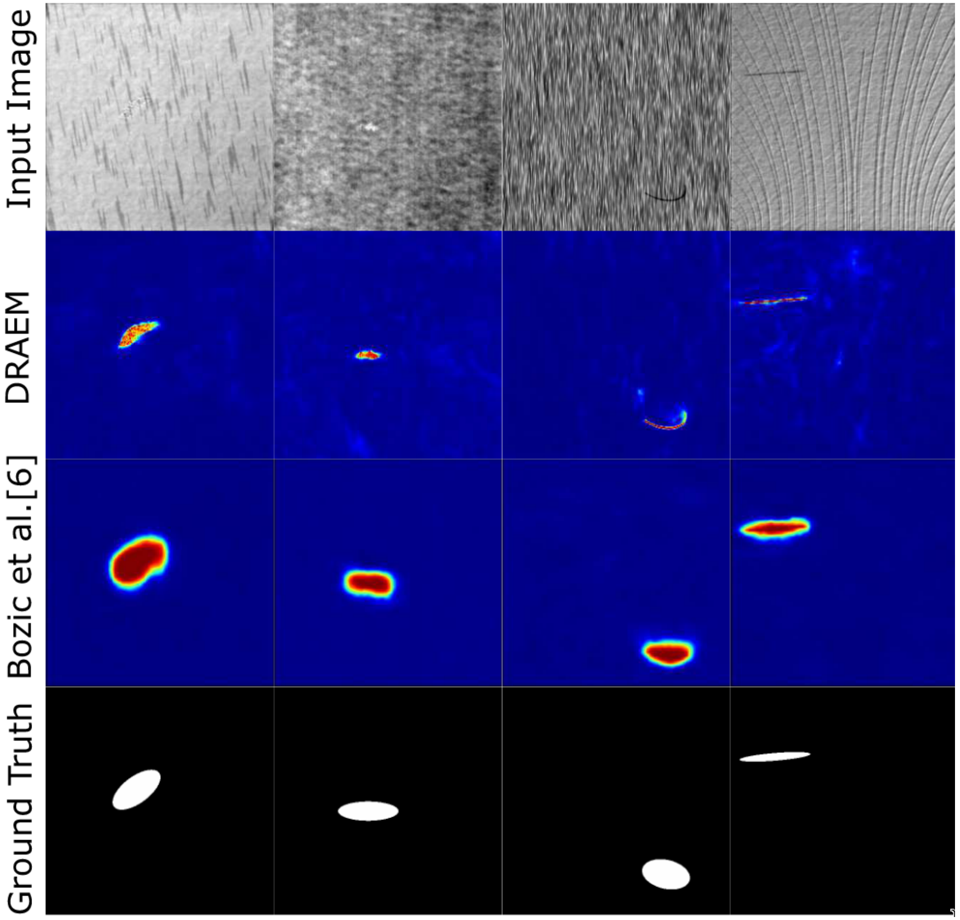

Figure: DRÆM vs supervised SOTA (Bozic et al.) on the DAGM dataset.

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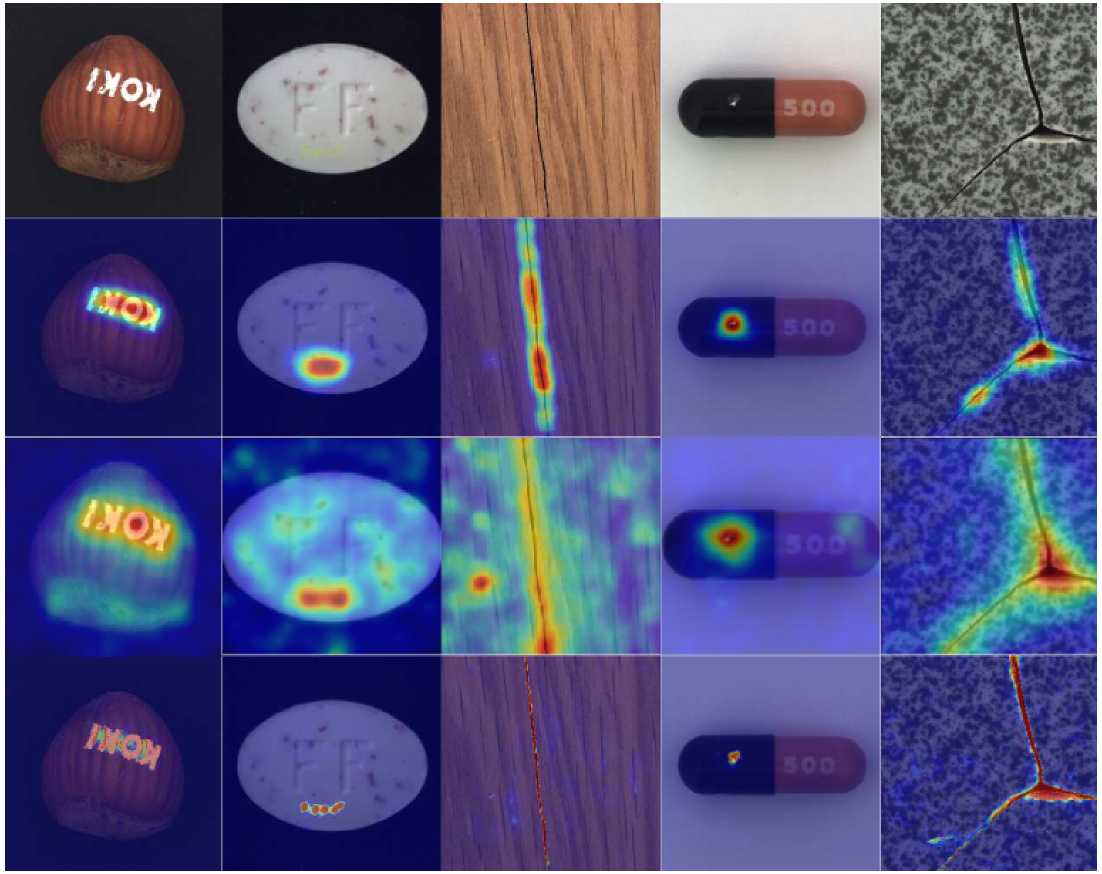

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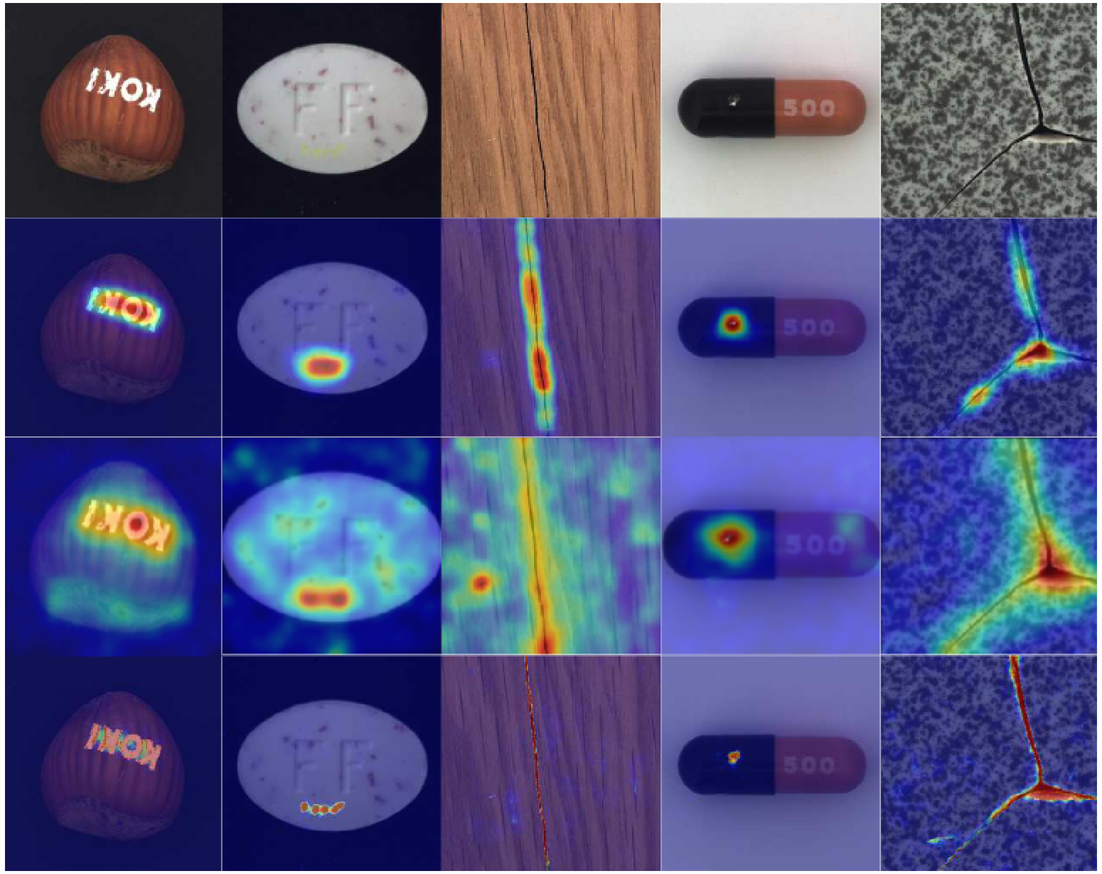

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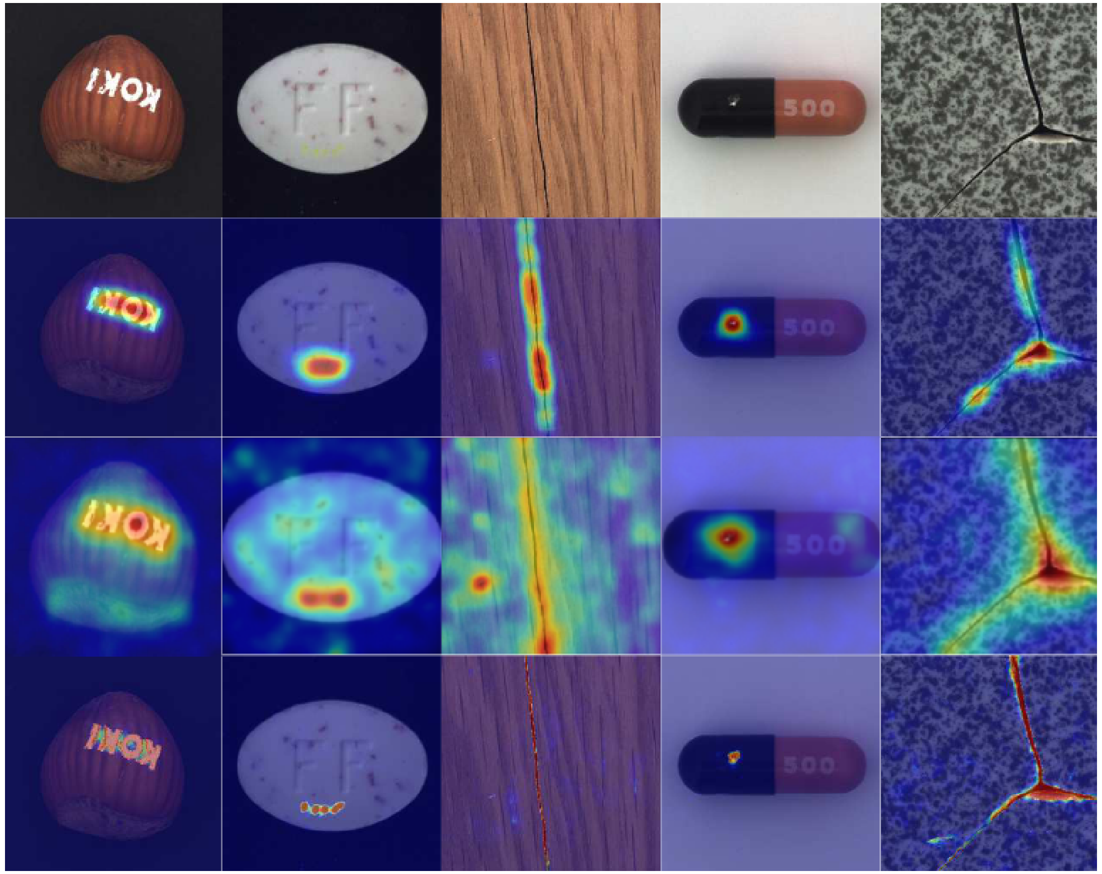

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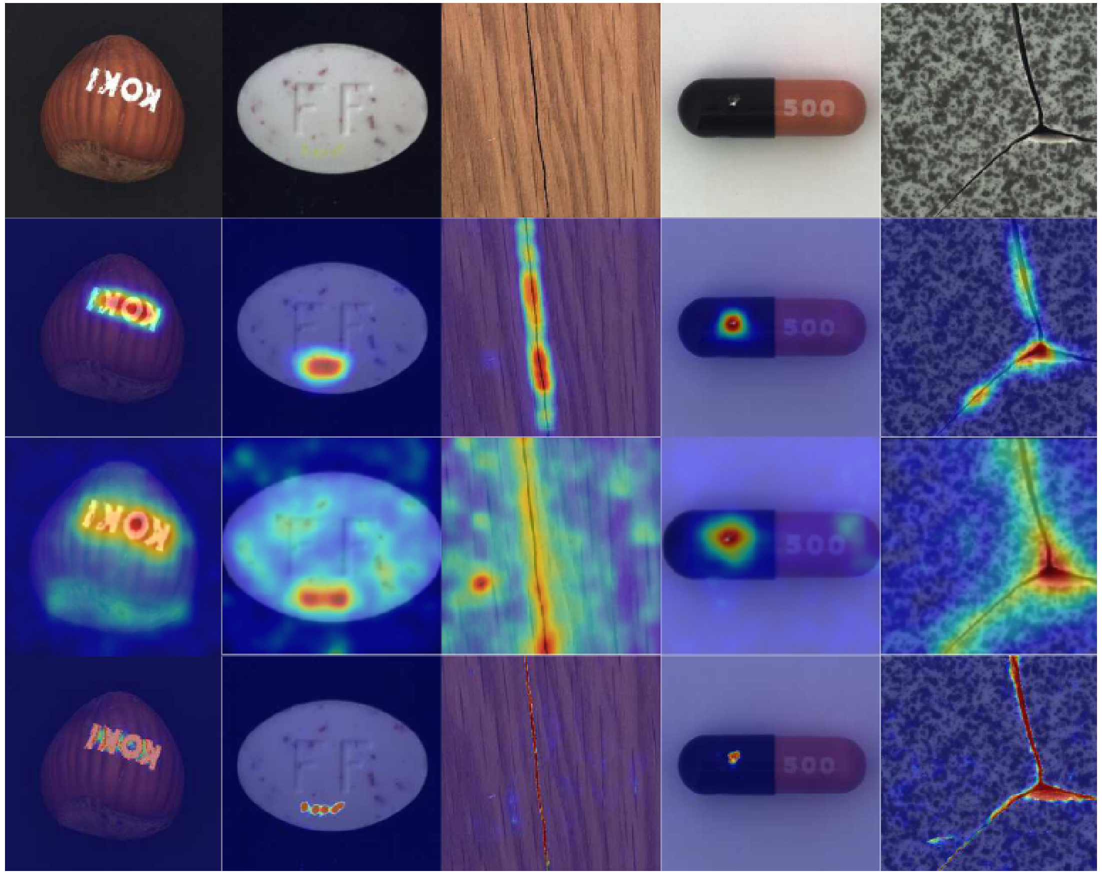

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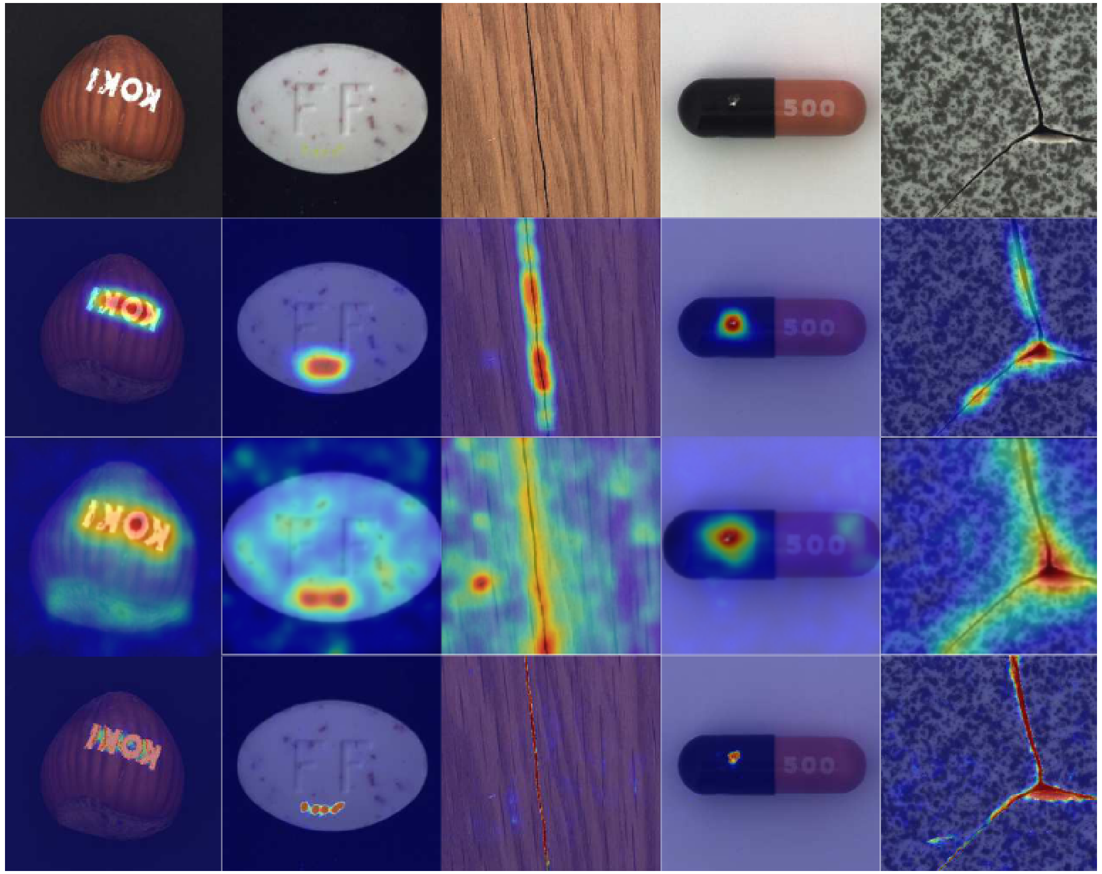

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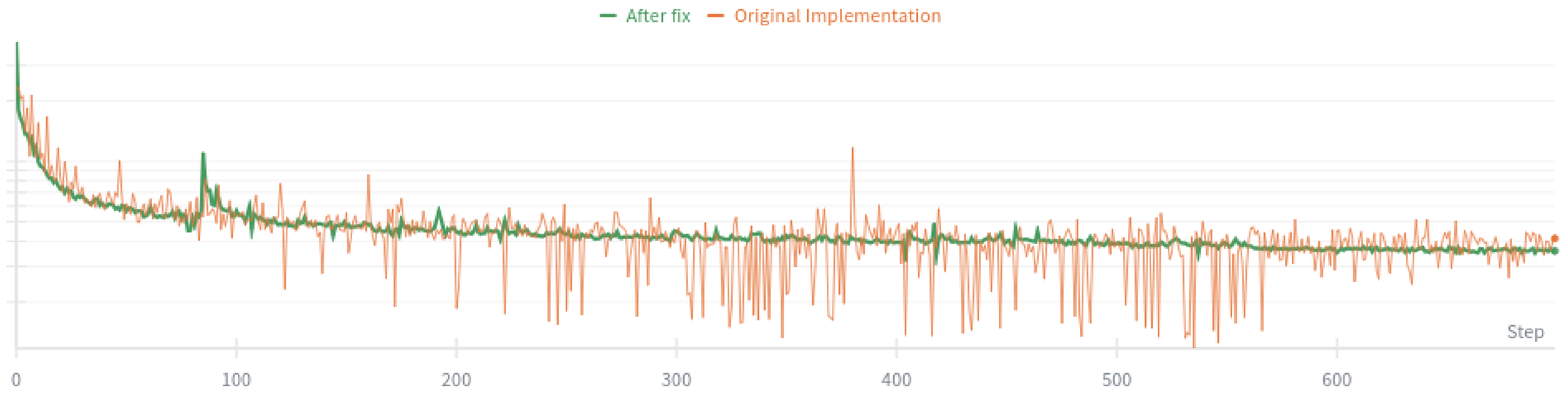

Figure: Loss smoothing by weighting the last batch of each epoch.

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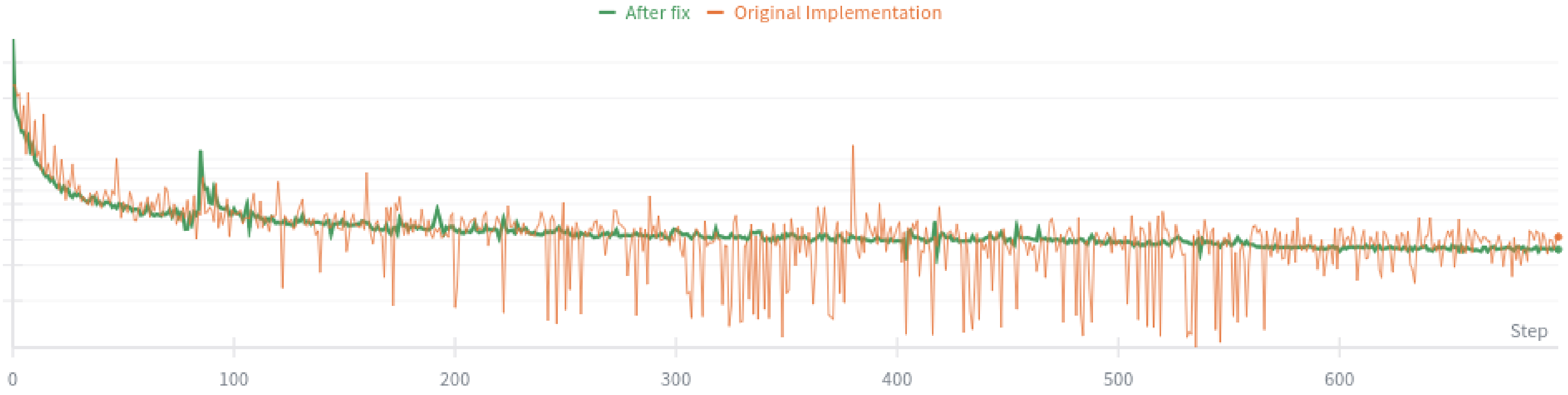

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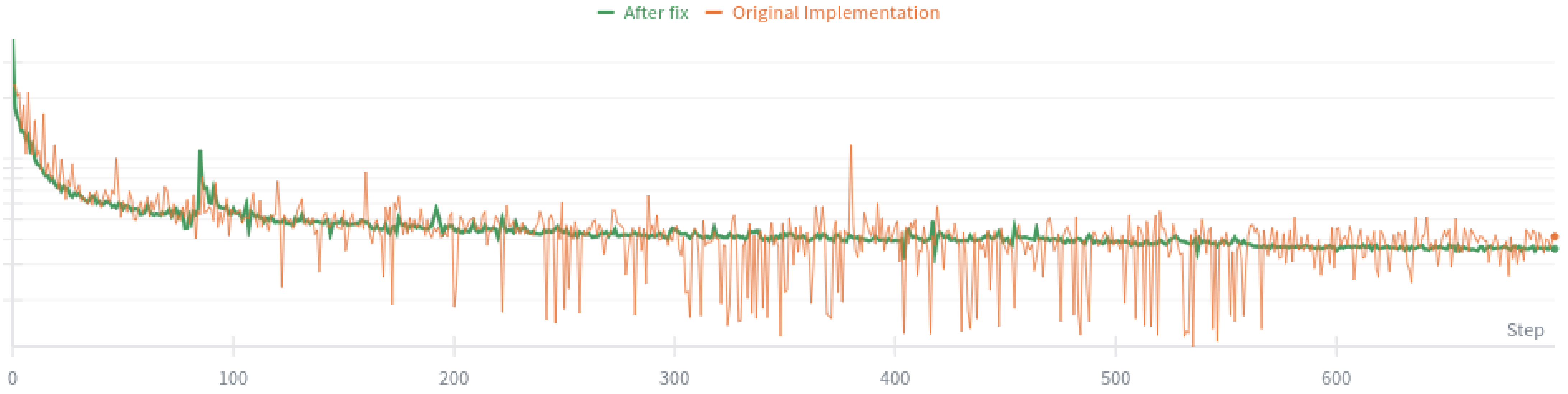

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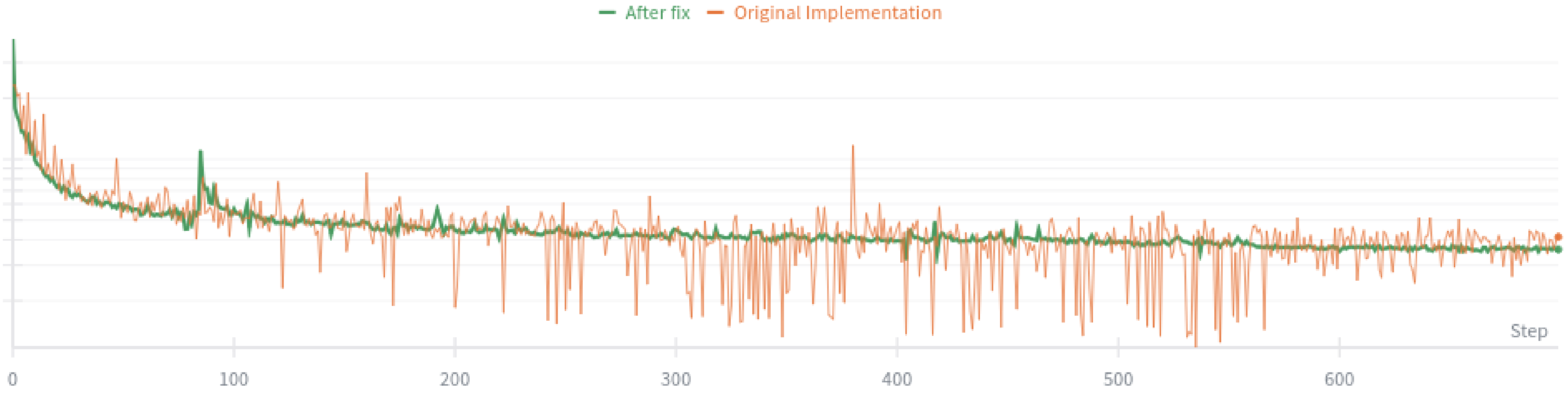
- ✓ Download script for MVTec and DTD datasets



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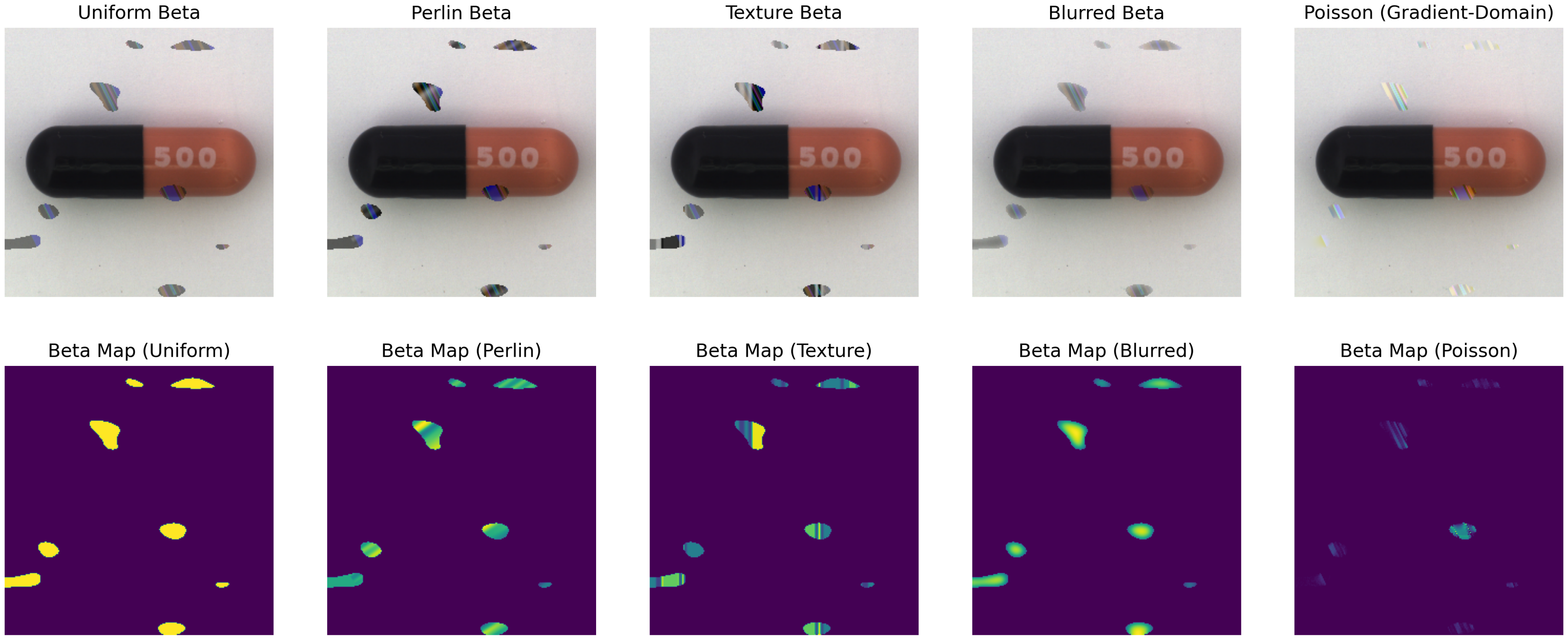

Figure: Different blending methods (from left to right): uniform β, Perlin-β, texture-β, blurred β, Poisson Image Interpolation.

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DRÆM-Texture	98.0%	97.2%	69.1%
DRÆM-Perlin	98.2%	97.3%	69.5%
DRÆM-Poisson	<u>99.1%</u>	<u>97.8%</u>	<u>71.6%</u>

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DRÆM-Poisson	99.1%	97.8%	71.6%
DRÆM-Poisson+Perlin	**99.2%**	**98.0%**	**73.6%**

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Thank you for your attention.